THARUSHI WITHARANA

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Summary

Software engineer, BSc.(Hons) Engineering graduate with **2+ years of industry experience building enterprise software solutions and agentic AI solutions**. Passionate about solving complex problems through clean, maintainable software and delivering high-quality solutions.

Skills

Languages: C/C++, Java, Python, JavaScript/TypeScript, SQL, Erlang
Frontend: React.js, HTML, CSS, Tailwind CSS, ShadCN, Ant Design
Backend: Express.js, Node.js, Spring Boot, FastAPI, Flask
APIs: REST, WebSocket, SSE
Security: Keycloak (OAuth2/OIDC)
AI/ML: LLM APIs, LangChain, LangGraph, Google ADK, RAG, Azure AI Search, MCP
Databases & Caching: MySQL, MariaDB, MongoDB, DynamoDB, Redis
Cloud & DevOps: AWS, DigitalOcean, Linux, Docker, Nginx, PM2
Concepts & Foundations: Data Structures and Algorithms, Database Management Systems, Software Architecture, Computer Vision and Image Processing, Machine Learning, Deep Learning, Object-Oriented Design Patterns and Principles

Experience

Global Wavenet International (Pvt) Ltd — Nov 2024 – Present

Associate Software Engineer — *Apr 2025 – Present*

- Contributed to the development of **10+ projects**, including 8 agentic AI applications, 1 large-scale SaaS product, and several full-stack web applications.
- **Developed and delivered a $16K change request within 2 weeks** for Claro, Clombia. Technical Visit, an agentic AI voice agent for call center agent replacements using Google ADK and FastAPI. CR contained application security related enhancements
- Delivered **full-stack application development** across three call center products: WFM, Agent 360, and Auto Pilot. Built the frontend and backend of WFM, a workforce management system for call center agents, using React, FastAPI, and MySQL, with database migrations managed through Alembic and authentication through CAM. Fixed frontend and backend bugs in Agent 360, a call center operations platform for queue management, customer engagement history, and skill-based agent management. Developed a WebRTC signaling server and a React frontend for customer demos of Auto Pilot, a customer video call application.
- Developed and enhanced the Intent Router Agent and the Intent Router Agent with Compose, **Google ADK-based agentic voice agents** that match customer intents with and without Compose integration. Built two additional voice agents, the Roaming Information Agent and the Internet and Device Configuration Agent, using Google ADK and FastAPI, and calculated AI voice agent costs for customer proposals.
- Migrated the frontend of Compose, a large-scale SaaS platform for building call flows, messaging, and telecommunication service integrations on a drag-and-drop canvas, from Angular 7 to React 19. **Reduced loading time from several minutes to a few seconds** and improved the UI for a better customer experience.
- Built **AI agents for Agentic Campaign Manager covering campaign creation, segmentation, and natural-language database retrieval**, orchestrating three specialized agents through a LangGraph supervisor agent that guides users through campaign creation. Developed and enhanced the React UI for customer demonstrations.
- Enhanced the features and UI of AI-Troubleshooter v2.0.0 and InsightDB v2.0.0 using the MERN stack, and upgraded InsightDB's AI framework from LangChain to LangGraph. Deployed both applications on local servers and integrated them into AI-Ops, a desktop application. Contributed to **full-stack and agentic AI development**, including desktop application and CLI tool development using OCLIF.

Software Engineering Intern — *Nov 2024 – Apr 2025*

- Contributed to **4 projects**, including 2 large-scale enterprise SaaS web applications and 2 AI research and development (R&D) projects.
- Built AI-Troubleshooter v1.0, an agentic AI application that helps operations engineers resolve L1 and L2 tickets by connecting to internal servers via SSH, achieving **84.4% accuracy**. Developed it with React, Node.js, Express.js, and Keycloak authentication; handled database integration and R&D on AI and RAG integration; and deployed it with Nginx and PM2, and later with Docker.

- Built InsightDB v1.0, an agentic AI application that retrieves data from SQL and NoSQL databases using natural language and visualizes the results, **reducing report generation time by 66.7%–80%** for an enterprise business analyst team. Developed it with React, Node.js, Express.js, Keycloak authentication, and LangChain, and deployed it on a local server with Nginx and PM2.
- Fixed bugs in Prime and Campaign Manager deliverables.

Department of Electrical and Information Engineering, FOE, UOR **Jan 2024 – Jan 2025**

Student Instructor *Galle, Sri Lanka*

Lightway IT (Pvt) Ltd **Sep 2023 – Jan 2024**

Software Engineering Intern *Remote*

- Developed a **RAG-based customer assistance chatbot** using LangChain and OpenAI GPT, reducing response time and improving customer satisfaction. Integrated the chatbot with WhatsApp, Instagram, Facebook, and Telegram webhooks through a REST endpoint on AWS Lambda, and built the management backend using Node.js and Express.js with DynamoDB and MySQL.

Projects

Tea Leaf Disease Detection via Machine Learning | *ML, DL, Flutter, Flask, Firebase* **Jan 2024 – Nov 2024**

- Relates to final year research project and published and presented at the APITT International Conference 2024
- Achieved 94% of testing accuracy on a collected novel dataset of 1,432 images. Choosen best suited model to deploy on mobile app by comparing 4 machine learning algorithm results and 5 deep learning algorithm results on kaggle dataset.
- Implemented the machine learning and deep learning models. Collected and processes raw images. Developed mobile app using Flutter, authentication using firebase and backend using Flask. Deployed separate backend on GCP and Huggingface for online disease detection

FuelPass - Fuel Management System | *Java Spring Boot, Microservices, Flutter, REST API* **Aug 2024 – Nov 2024**

- Contained four web portals and one mobile application for users, fuel station owners, fuel station workers, and system admins to manage the fuel quota allocated to each vehicle, similar to the system used in Sri Lanka during the pandemic
- Developed the mobile application and its related microservices using Flutter and Spring Boot with inter-service communication

FYPMarks - Web Application for Entering Final Year Project Marks | *MERN Stack, REST API* **Jan 2023 – Sept 2023**

- A role-based MERN stack platform for managing final year university projects across students, evaluators, HODs, and administrators, with structured workflows and access control. Implemented CSV bulk registration, three-phase evaluation process, group and individual grading, Cloudinary-based media handling, and admin analytics with automated mark finalization.
- Developed role based auth using Keycloak and developed fronend using React, backend for the dynamically changing evaluation forms using Node.js, Express.js and MongoDB Deployed the production-grade architecture with React frontend on S3, Node.js/Express.js backend on EC2 instance, MongoDB Atlas as the managed database, IAM roles for least-privilege access

Education

University of Ruhuna, Galle, Sri Lanka **March 2020 – March 2025**

BSc. Eng (Hons) in Electrical and Information Engineering — Second Class Upper Division (GPA 3.46)

Certifications and Badges

Microsoft Learn: Cloud Skills Challenge — Azure Fundamentals (AZ-900)
HackerRank Badges: C++ Silver Badge, Java Silver Badge
LinkedIn Learning: AWS Exam Prep, Learning Ansible, Learning Kubernetes, MERN Essential Training

Leadership and Extracurricular Activities

Blog Article Writing: Chatbot Development with the OpenAI Assistant API: A Step-by-Step Guide
Sports: Faculty team player in volleyball, elle, and cricket, with school-level sports achievements
Leadership: Assistant girls' batch representative, public relations community member of IEEE ComSoc, and Sports captain (Girls)
Clubs: Student member of IESL, Gaveshakayo, CEDS, and the LEO Club

References

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